

# ITAI 1371 — Module 2 Reflection Journal

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## Module 2: Tools Used in Machine Learning

This lab changed my understanding of machine-learning tools because I could see how several tools work together as one workflow. Before working with the Iris dataset, pandas, NumPy, and Matplotlib could seem like separate programming topics. Using them in sequence made the connection clearer: pandas organized the data, NumPy performed numerical calculations, and Matplotlib turned numerical relationships into a visual pattern. The workflow showed me that learning a tool is not only about memorizing commands; it is about understanding what problem each tool helps solve.

One important insight came from comparing the table of Iris measurements with the scatter plot. The numerical output contains many values, but the visualization makes grouping and overlap easier to recognize. The plot showed a clear separation for setosa while versicolor and virginica overlapped more. That helped me understand why visualization is useful before building a machine-learning model: it can reveal structure, overlap, and questions that may not be obvious from raw numbers alone.

The statistical calculations also helped connect programming with concepts I already know from mathematics. Calculating the mean and standard deviation with NumPy was more meaningful when I could relate the results to the actual sepal-length measurements. I learned that a mean gives a central value, while standard deviation gives information about how spread out the observations are. This made the code feel less like a formula to memorize and more like a practical way to summarize data.

A challenge in this lab was keeping track of the relationship between the dataset structure and the code. For example, the Iris dataset begins with numeric target values, while the DataFrame uses readable species names. Following that transformation helped me see why data preparation matters: readable and well-organized data makes later analysis easier and less error-prone. I also learned to pay attention to verification statements and output messages because they provide quick evidence that a calculation or task worked as intended.

The lab also changed how I think about reproducibility. Running the notebook from the beginning is important because later cells depend on variables created earlier. Re-running the cells helped reinforce that a successful notebook should not depend on hidden or accidental state. This is especially important in machine learning, where another person should be able to follow the same workflow and understand how the results were produced.

For further learning, I would like to investigate whether petal measurements separate the three Iris species more clearly than sepal measurements. I would also like to experiment with additional visualizations and standardization. These questions connect the basic tools from this lab to the larger machine-learning workflow I will use in later modules.

**Overall reflection:** The most useful lesson was learning how the tools complement one another. Instead of viewing Python libraries as isolated requirements, I now see them as parts of a repeatable data-analysis process: load and organize data, calculate useful summaries, visualize patterns, verify results, and document the work clearly.